

Task 7 – Exploratory Data Analysis (EDA)

Topics Covered

Understanding dataset structure and summary statistics
Identifying missing values and outliers
Visualizing distributions and relationships between variables

Program 1 – EDA with Pandas and Seaborn

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load dataset
data = sns.load_dataset("titanic")

# Summary statistics
print(data.describe())

# Check for missing values
print(data.isnull().sum())

# Visualize age distribution
sns.histplot(data['age'], bins=30, kde=True)
plt.title("Age Distribution")
plt.show()

# Correlation heatmap
sns.heatmap(data.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

Key Takeaways

Learned how to perform EDA using Pandas and Seaborn.
Understood data distributions and detected missing values.
Discovered correlations and outliers visually.
Improved understanding of dataset structure before modeling.